Lecture 1: Practice Problems — Answers and Hints

Linear Algebra

October 18, 2026

Number Sets and Fields

1.1. $\frac{7}{3}, 0.25 \in \mathbb{Q}, \mathbb{R}, \mathbb{C}$; $\sqrt{49} = 7$, $\pi - \pi = 0$ and $(1+i)(1-i) = 2$ lie in all four sets; $\sqrt{12} = 2\sqrt{3} \in \mathbb{R}, \mathbb{C}$; $2 + 3i \in \mathbb{C}$ only.

1.2. (a) Additive inverses: $-1 \notin \mathbb{N}$. (b) There is no additive identity, since $0 \notin \mathbb{R}_{>0}$. (c) Not closed under addition: $1 + 1 = 2$. (d) There is no multiplicative identity, since $1 \notin 2\mathbb{Z}$.

1.3. (a) $(a + b\sqrt{2})^{-1} = \frac{a}{a^2-2b^2} - \frac{b}{a^2-2b^2}\sqrt{2}$; if $a^2 = 2b^2$ with $(a, b) \neq (0, 0)$, then $\sqrt{2} = |a/b|$ would be rational. (b) No: 2 has no inverse, since $\frac{1}{2} = a + b\sqrt{2}$ forces $b = 0$ and $a = \frac{1}{2} \notin \mathbb{Z}$.

Vector Spaces

2.1. (a) Only $1 \odot v = v$ fails. (b) Only $(a+b) \odot v = a \odot v + b \odot v$ fails: the second coordinates are y and $2y$.

2.2. The zero vector is 1, and the additive inverse of u is $\frac{1}{u}$. For example, $(a + b) \odot u = u^{a+b} = u^a u^b = (a \odot u) \oplus (b \odot u)$.

2.3. (a) The zero vector is the zero function $x \mapsto 0$. (b) No: the zero function is not in the set (and the set is not closed under addition).

Bases and Dimension

3.1. Dependent: $(1, 2, 1) - (2, 1, 0) + (1, -1, -1) = 0$. The span has basis $\{(1, 2, 1), (2, 1, 0)\}$ and dimension 2.

3.2. $\dim U = 2$, with basis $(1, 0, -2, 1)$, $(0, 1, -1, 0)$.

3.3. Substitute $x = t + 1$: $[x^2]_{\mathcal{B}} = (1, 2, 1)$ and $[3 + 2x + x^2]_{\mathcal{B}} = (6, 4, 1)$.

Linear Transformations

4.1. (a) Linear. (b) Not linear: $T(0, 0) \neq (0, 0)$. (c) Not linear: $T(2v) = 4T(v)$ in the first coordinate. (d) Linear. (e) Not linear: $T(-1) = 1 \neq -T(1)$. (f) Linear.

4.2. $\ker(T) = \operatorname{span}\{(1, -1, 1)\}$ and $\operatorname{im}(T) = \{(a, b, c) : c = a - b\} = \operatorname{span}\{(1, 0, 1), (1, 1, 0)\}$. T is neither injective nor surjective.

4.3. (a) $T(e_1) = e_2$, $T(e_2) = e_1$; bijective; the F is mirrored. (b) $\ker(T)$ is the y-axis and $\operatorname{im}(T)$ is the x-axis; the square collapses onto the segment from $(0,0)$ to $(1,0)$. (c) Bijective; the square becomes the parallelogram with vertices $(0,0), (1,0), (3,1), (2,1)$. (d) Rotation by 180°; bijective; the F is not mirrored. Only (a) mirrors the F.

4.4. $\ker(D)$ is the set of constant polynomials, and $\operatorname{im}(D) = \mathbb{R}[x]_{\leq 2}$. D is neither injective nor surjective. $D \circ D \circ D \circ D = 0$.

4.5. (a) If $f(a) = f(b)$, then $g(f(a)) = g(f(b))$, so $a = b$. (b) $f : \{1\} \to \{1,2\}$, $f(1) = 1$, and $g : \{1,2\} \to \{1\}$. (d) A is injective and B is surjective.

4.6. (a) For example, $f(x) = x + 1$ and $g(x) = 2x$. (c) With $f(x) = x^2$ and $g = h = \mathrm{id}$: $f(2x) = 4x^2 \neq 2x^2$. (d) $c(g(x) + h(x)) = cg(x) + ch(x)$.

4.7. (a) $T(a_0 + a_1x + a_2x^2 + a_3x^3) = (a_0, a_1, a_2, a_3)$. (b) $\log : V \to \mathbb{R}$, since $\log(uv) = \log u + \log v$ and $\log(u^c) = c \log u$. (c) Yes; no (dimensions 2 and 3); yes.

Subspaces

5.1. (a) yes; (b) no; (c) no; (d) no; (e) yes; (f) yes.

5.2. (b) The x-axis and the y-axis in $\mathbb{R}^2$: $(1,0) + (0,1) = (1,1)$ lies in neither. (c) If $u \in U \setminus W$ and $w \in W \setminus U$, then $u + w$ lies in neither U nor W.

5.3. It is the kernel of $p \mapsto p(1)$. A basis is $x - 1,\ x^2 - 1,\ x^3 - 1$, so the dimension is 3.

Matrices

6.1. (a) $Ax = 0$. (b) $AA^\top = \begin{pmatrix} 5 & 1 \\ 1 & 11 \end{pmatrix}$ and $A^\top A = \begin{pmatrix} 2 & -3 & 1 \\ -3 & 9 & 3 \\ 1 & 3 & 5 \end{pmatrix}$, both symmetric.

6.2. AB: 2×3; BA: undefined; AC: 2×2; CA: 3×3; $A + C^\top$: 2×3; $A^\top + C$: 3×2; ABC: 2×2.

6.3. $FR = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ is the reflection in the line $y = -x$, and $RF = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is the reflection in the line $y = x$. So $FR \neq RF$.

6.4. (b) $\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$ and $\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$.

6.5. (c) $M(1+i)^{10} = 2^5 R_{10\pi/4} = 32R_{\pi/2} = \begin{pmatrix} 0 & -32 \\ 32 & 0 \end{pmatrix} = M(32i)$, so $(1+i)^{10} = 32i$.

6.6. The first is in RREF. The second is in REF, but not RREF. The third and fourth are not in REF (the pivots move left; a zero row lies above a nonzero row). The fifth is not in REF with our convention, since its first pivot is 2.

6.7. $\operatorname{RREF}(P) = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, and $\operatorname{rank}(P) = 2$.

6.8. The RREF has rows $(1,0,5,8)$ and $(0,1,-1,-2)$, so the solutions are

$$(w,x,y,z) = y\,(-5,1,1,0) + z\,(-8,2,0,1), \qquad y,z \in \mathbb{R}.$$

The solution $(0,2,-8,5)$ is the case $y=-8$, $z=5$; it is not a multiple of $(-5,1,1,0)$. The dimension is 2.

6.9. $A^{-1} = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$.

6.10. $A^{-1} = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 1 \end{pmatrix}$ and $x = (1,-2,4)^\top$.

6.11. (a) $x = (1,-2,1)^\top$. (b) row 1 $-$ 2·row 2 + row 3 $= 0$, so row operations produce a zero row.

6.12. Swap the rows, multiply row 2 by $\frac{1}{2}$, subtract 3 times row 2 from row 1: $E_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $E_2 = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$, $E_3 = \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix}$. Then $A^{-1} = E_3E_2E_1 = \begin{pmatrix} -\frac{3}{2} & 1 \\ \frac{1}{2} & 0 \end{pmatrix}$ and $A = E_1^{-1}E_2^{-1}E_3^{-1}$.

6.13. (a) $A = A^{-1}A^2 = A^{-1}A = I$. (b) $\ker(T_P) = \operatorname{span}\{(1,-1)\}$ and $\operatorname{im}(T_P)$ is the x-axis: T_P projects onto the x-axis along lines parallel to $(1,-1)$. (d) If $v = Aw$ and $Av = 0$, then $v = Aw = A^2w = Av = 0$.

6.14. $N^2 = \begin{pmatrix} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $N^3 = 0$, and $(I-N)^{-1} = I + N + N^2 = \begin{pmatrix} 1 & 1 & 5 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$.

6.15. (a) $\begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$, with determinant 2. (c) $p = 1 - x^2$ and $q = 2 + 2x + x^2$.

6.16. (a) 3, basis $e_{11}, e_{22}, e_{12} + e_{21}$; (b) 1, basis $e_{12} - e_{21}$; (c) 6; (d) 10; (e) 6; (f) 5.

6.17. (a) Yes. (b) No: $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}$ is not symmetric. (c) No: $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ square to I_2, but their product squares to $-I_2$. (d) Yes. (e) Yes; there are $n!$ of them, and each one is orthogonal.

6.18. $\begin{pmatrix} 1 & -2 & 5 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{pmatrix}$. For an upper triangular A with nonzero diagonal entries, the Gauss–Jordan method only needs to scale rows and to subtract multiples of lower rows from higher rows, and these operations keep the right block upper triangular.

6.19. $Q = R_\theta$ (rotations, determinant 1) or $Q = R_\theta \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$ (reflections, determinant -1).

6.20. (a) $D_1 = 2$, $D_2 = 3$, $D_3 = 4$. (c) $D_n = n + 1$.

6.21. $\det(A) = -1$, so A is invertible.

6.22. -6, 18, 54, $-\frac{3}{2}$, 2; and $\det(C) = -2$.

6.23. (a) Subtract row 1 from rows 2 and 3, and take out the factors $b-a$ and $c-a$. (b) The determinant is nonzero, so the map $p \mapsto (p(a), p(b), p(c))$ is an isomorphism.

6.24. (a) $\lambda = 2$ and $\lambda = 5$; $\det(A - 3I_3) = -16 \neq 0$. (b) $x = (-8, 1, 1)$ for $\lambda = 2$, and $x = (-1, -1, 1)$ for $\lambda = 5$. (c) Every row sums to 11, so $A(1,1,1)^\top = 11(1,1,1)^\top$.

6.25. (a) $\det(A) = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120$; yes; yes. (b) $(0, 0, 0, 0, 25)$. (c) $A - I_5$ and $A - 2I_5$ are triangular with a 0 on the diagonal, so their determinants are 0. Take $x = e_1$ and $y = (2, 1, 0, 0, 0)^\top$.

6.26. (a) $\frac{1}{2}\left|\det \begin{pmatrix} 3 & 1 \\ 1 & 4 \end{pmatrix}\right| = \frac{11}{2}$. (b) $\det = 7$: the areas are 7 and 7π. (d) Only the reflection, with determinant -1. The rotation by 180° has determinant $+1$.

6.27. $M^{10} = \begin{pmatrix} 89 & 55 \\ 55 & 34 \end{pmatrix}$, and $89 \cdot 34 - 55^2 = 1$. Cassini follows from $\det(M^n) = \det(M)^n = (-1)^n$.

6.28. (a) For example, $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. $\operatorname{tr}(AB - BA) = 0 \neq 2 = \operatorname{tr}(I_2)$. (b) $\operatorname{tr}(A^\top A)$ is the sum of the squares of all entries. (c) $\operatorname{tr}(Ae_{ji}) = A_{ij}$.

6.29. (a) 6. (For 3×3 matrices with entries in $\{0, 1\}$, a computer finds that 174 of the 512 are invertible.) (b) If A and A^{-1} have integer entries, then $\det(A)\det(A^{-1}) = 1$ is a product of two integers. Conversely, use the cofactor formula $A^{-1} = \frac{1}{\det(A)} C^\top$.

Null Space, Column Space and Rank

7.1. $\operatorname{RREF}(B)$ has rows $(1, 2, 0, 3)$, $(0, 0, 1, 2)$, $(0, 0, 0, 0)$, and $\operatorname{rank}(B) = 2$. Bases: $\operatorname{null}(B)$: $(-2, 1, 0, 0)$, $(-3, 0, -2, 1)$; $\operatorname{col}(B)$: $(1, 2, 1)$, $(0, 1, 1)$; $\operatorname{row}(B)$: $(1, 2, 0, 3)$, $(0, 0, 1, 2)$. Indeed, $2 + 2 = 4$.

7.2. (a) $5 - 3 = 2$. (b) Not surjective, since the rank is at most 3; it can be injective. (c) No: $\dim \operatorname{null}(A) \geq 6 - 4 = 2$. (d) 3.

7.3. (a) The 7×8 matrix has a nonzero null vector $(a_1, \ldots, a_4, b_1, \ldots, b_4)$; then $\sum a_i v_i = -\sum b_j w_j$ is a nonzero vector in $V \cap W$. (c) $\operatorname{span}\{e_1, e_2, e_3\}$ and $\operatorname{span}\{e_4, e_5, e_6\}$.

7.4. (b) $\operatorname{rank}(AB) \leq \operatorname{rank}(A) \leq 2 < 3$.

7.5. (a) $x^\top A^\top A x = \sum_i (Ax)_i^2$. (b) Use the rank–nullity theorem. (c) $\operatorname{null}(A^\top A) = \operatorname{null}(A) = \{0\}$.

Dual Vector Spaces

8.1. (a) Yes, $a = (2, 0, -1)$. (b) No. (c) Yes. (d) No: $\det(2A) = 4\det(A)$. (e) Yes. (f) No.

8.2. (a) $P = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ has the inverse $\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$, so $b^1(x) = x_1 - x_2$ and $b^2(x) = -x_1 + 2x_2$. (b) $\varphi = \varphi(b_1)b^1 + \varphi(b_2)b^2 = 7b^1 + 4b^2$.

8.3. (a) $L_0 = \frac{(x-1)(x-2)}{2}$, $L_1 = -x(x-2)$, $L_2 = \frac{x(x-1)}{2}$. (b) $p = L_0 + 2L_1 + 5L_2$, since $p(0) = 1$, $p(1) = 2$, $p(2) = 5$. (c) $I(L_0) = \frac{1}{3}$, $I(L_1) = \frac{4}{3}$, $I(L_2) = \frac{1}{3}$, so $\int_0^2 p(x)\,dx = \frac{1}{3}(p(0) + 4p(1) + p(2))$.

8.4. (a) $(2,-1)A = (2,-1)\begin{pmatrix}1 & 1 & 0\\ 0 & 1 & -1\end{pmatrix} = (2,1,1)$. (b) $\psi(T(v)) = \psi(3,-1) = 7$ and $(2,1,1)v = 7$. (c) $A^\top$.

8.5. (a) The rows a with $a_1 + 2a_2 = 0$ and $a_2 + a_3 = 0$, spanned by $(2,-1,1)$; $2+1=3$. (b) $(1,0,-1,0)$ and $(0,1,0,-1)$.

8.6. (a) $y = (2,-1)$: $yA = 0$ and $yb = -1$. (b) The rows with $yA = 0$ are those with $y_1+2y_2+3y_3 = 0$, spanned by $(-2,1,0)$ and $(-3,0,1)$; so the system is solvable if and only if $b_2 = 2b_1$ and $b_3 = 3b_1$, i.e., b is a multiple of $(1,2,3)^\top$.

8.7. $\iota(e_1)(\varphi) = \varphi(e_1) = a_1$: the functional "first entry of the row". If $\iota(v) = 0$, then $\varphi(v) = 0$ for all rows, in particular for $e_1^\top$ and $e_2^\top$, so $v_1 = v_2 = 0$.

Quotient Spaces and the Isomorphism Theorems

9.1. (a) Yes, since $(2,1)^\top \in U$; yes, since $(4,2)^\top \in U$; no, since $(1,2)^\top \notin U$. (b) $v - w \in U$ if and only if $v_1 - 2v_2 = w_1 - 2w_2$; the representative on the x_1-axis is $(c,0)^\top$. (c) $x + U \mapsto x_1 - 2x_2$.

9.2. (a) 3, so $\mathbb{R}^5/U \cong \mathbb{R}^3$. (b) $9-6=3$; it is isomorphic to Skew_3 (Problem 9.3). (c) The map $p \mapsto (p(1), p(2))$ is surjective with kernel U, so $\mathbb{R}[x]_{\le 4}/U \cong \mathbb{R}^2$ and $\dim U = 3$.

9.3. (a) T is surjective with kernel P, so $\mathbb{R}^3/P \cong \mathbb{R}$. (b) $p \mapsto (p(0), p(1))$ is surjective onto $\mathbb{R}^2$ with kernel U, so the quotient is $\cong \mathbb{R}^2$ and $\dim U = 2$ (spanned by $x(x-1)$ and $x^2(x-1)$). (c) $\ker(T) = \mathrm{Sym}_n$ and $\mathrm{im}(T) = \mathrm{Skew}_n$ (every skew-symmetric K equals $T(K/2)$), so $\mathbb{R}^{n\times n}/\mathrm{Sym}_n \cong \mathrm{Skew}_n$.

9.4. (a) $\dim(V \cap W) = 8 - \dim(V+W) \ge 1$, so the possible values are $1,2,3,4$; for example, $\mathrm{span}\{e_1,\dots,e_4\} \cap \mathrm{span}\{e_4,\dots,e_7\} = \mathrm{span}\{e_4\}$. (b) $U \ne W$, so $U + W = \mathbb{R}^4$ and $\dim(U \cap W) = 3+3-4 = 2$. Adding and subtracting the two equations gives $x_1 + x_3 = 0$ and $x_2 + x_4 = 0$, so a basis is $(1,0,-1,0)^\top$, $(0,1,0,-1)^\top$.

9.5. $V/U \cong \mathbb{R}^3$ (keep x_1,x_2,x_3), $W/U \cong \mathbb{R}$ (keep x_3), $V/W \cong \mathbb{R}^2$ (keep x_1,x_2); indeed $3-1=2$.

9.6. (a) The factor e^c cancels. (b) Both are approximately $(0.090, 0.245, 0.665)$; e^{1001} overflows in floating-point arithmetic, so we compute with the representative $(1001,1002,1003) - 1003\cdot\mathbf{1} = (-2,-1,0)$ instead. (c) Yes: $\frac{\mathrm{softmax}(z)_i}{\mathrm{softmax}(z)_j} = e^{z_i - z_j}$, so equal outputs give $z_i - z_j = w_i - w_j$ for all i,j, i.e., $z - w \in \mathrm{span}\{\mathbf{1}\}$.

9.7. The rows with $a_1 + a_2 + a_3 = 0$, e.g., spanned by $(1,-1,0)$ and $(1,0,-1)$.

Eigenvalues and Eigenvectors

10.1. (a) $p_A(\lambda) = \lambda^2 - 3\lambda - 4 = (\lambda-4)(\lambda+1)$; $E_4 = \mathrm{span}\{(2,3)^\top\}$ and $E_{-1} = \mathrm{span}\{(1,-1)^\top\}$. (b) $4 + (-1) = 3$ and $4\cdot(-1) = -4$. (c) $P = \begin{pmatrix}2 & 1\\ 3 & -1\end{pmatrix}$, $D = \mathrm{diag}(4,-1)$, $P^{-1} = \frac{1}{5}\begin{pmatrix}1 & 1\\ 3 & -2\end{pmatrix}$. (d) $A^k = \frac{1}{5}\begin{pmatrix}2\cdot 4^k + 3(-1)^k & 2\cdot 4^k - 2(-1)^k\\ 3\cdot 4^k - 3(-1)^k & 3\cdot 4^k + 2(-1)^k\end{pmatrix}$.

10.2. The eigenvalues are the diagonal entries 2, 3 and 1. The eigenspaces are spanned by $(1,-1,5)^\top$ (for 2), $(0,-2,1)^\top$ (for 3) and $(0,0,1)^\top$ (for 1). Diagonalizable, since the three eigenvalues are distinct.

10.3. The eigenvalue 2 has algebraic multiplicity 2, but geometric multiplicity $3-\operatorname{rank}(N-2I_3) = 1$; the eigenvalue 3 has both multiplicities equal to 1. Not diagonalizable.

10.4. (a) 1 on the x-axis, -1 on the y-axis. (b) 1 on $\operatorname{span}\{(1,1)\}$, 0 on $\operatorname{span}\{(1,-1)\}$. (c) Only 1, with eigenspace the x-axis. (d) No real eigenvalues; the complex eigenvalues are $\pm i$. (e) 2 on the x-axis, 3 on the y-axis. (f) Only -1, with eigenspace $\mathbb{R}^2$. All are diagonalizable except (c) and (d) (over $\mathbb{R}$).

10.5. (a) Yes (three distinct eigenvalues); yes (0 is not an eigenvalue). (b) 5, -8, $1+4+16 = 21$, $(-8)^3 = -512$. (c) A^{-1}: $-1, \frac{1}{2}, \frac{1}{4}$; A^2: $1, 4, 16$; $A + 2I_3$: $1, 4, 6$.

10.6. (a) If $Pv = \lambda v$, then $\lambda v = P^2 v = \lambda^2 v$, so $\lambda^2 = \lambda$. (b) If $v = Pw$, then $Pv = P^2 w = Pw = v$. Bases of $\operatorname{im}(T_P)$ and $\ker(T_P)$ together span $\mathbb{R}^n$ and are linearly independent, so they form a basis of eigenvectors. (c) Only "P is diagonalizable"; $P = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ is a counterexample to the other two.

10.7. (a) For the shear, $E_1 = \operatorname{span}\{e_1\}$, but for its transpose, $E_1 = \operatorname{span}\{e_2\}$. (c) $v^\top v = (Qv)^\top (Qv) = \lambda^2 v^\top v$. (d) $A^k v = \lambda^k v = 0$. (e) $P^{-1} I_2 P = I_2$ for every P.

10.8. (a) $M = \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix}$. (b) Eigenvalues 2 and -1, with eigenvectors $(2,1)^\top$ and $(-1,1)^\top$; $a_n = \frac{2^n - (-1)^n}{3}$. (c) $a_{10} = \frac{1024-1}{3} = 341$.

10.9. (a) $\lambda = 1$ with eigenvector $(4,3)^\top$, and $\lambda = 0.3$ with eigenvector $(1,-1)^\top$; the stationary distribution is $(\frac{4}{7}, \frac{3}{7})^\top$. (b) $x_k = (\frac{4}{7}, \frac{3}{7})^\top + \frac{3}{7}(0.3)^k (1,-1)^\top$; $x_2 = (0.61, 0.39)^\top$; $x_k \to (\frac{4}{7}, \frac{3}{7})^\top$.

10.10. (a) $(2,1)$, $(5,4)$, $(14,13)$, $(41,40)$; the ratio tends to 1, i.e., to the direction of the eigenvector $(1,1)$. (b) $x_0 = \frac{1}{2}(1,1)^\top + \frac{1}{2}(1,-1)^\top$, so $A^k x_0 = \frac{1}{2}(3^k + 1,\ 3^k - 1)^\top$.

10.11. (a) $p(\lambda) = (a-\lambda)^2 + b^2$. (b) $3 \pm 4i$; the matrix is $5R_\theta$ with $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$: $r = |3 \pm 4i| = 5$, and θ is the angle of $3 + 4i$.